

INDIA'S MISSION

Union Cabinet cleared the National Quantum Mission (NQM) in April 2023 with a ₹6,003 crore outlay over eight years

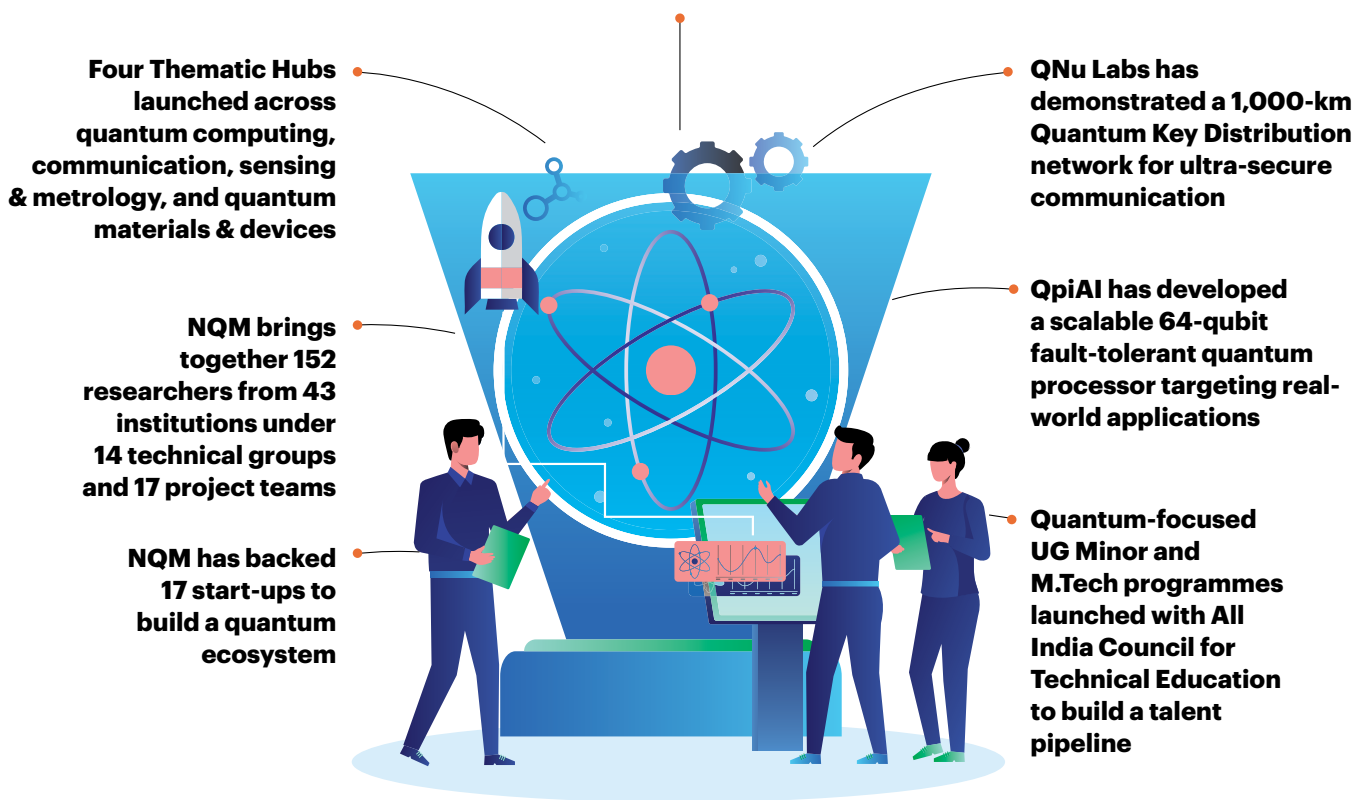


ILLUSTRATION BY **RAJ VERMA**

Mission (NQM) to build domestic capabilities in quantum computing, communications, and sensing. Can it be the leader in this frontier technology?

THE QUANTUM POWER

“Over the last year-and-a-half, we’ve seen rapid gains in quantum computing in terms of the technology stack and algorithmic advances in multiple application areas. Today quantum computing is being applied in a ‘quantum-centric supercomputing’ architecture, meaning that quantum processors (QPUs) work alongside GPUs and CPUs to tackle scientific challenges better than any single computing approach,” says Amith Singhee, Director, IBM Research India, and CTO, IBM India and South Asia.

The \$3.52 billion global quantum computing market is projected to cross \$20.20 billion over the next five years, as per Markets & Markets Research. But the race

is being driven less by immediate commercial returns and more by strategic implications. From cybersecurity and defence to intelligence gathering and technological sovereignty, governments are increasingly viewing quantum capabilities as a source of economic and geopolitical power.

India’s quantum ambitions, too, are no longer confined to research papers and laboratory experiments. “The country has moved from exploratory research to mission-mode execution. Our emphasis is on building sustainable and strategically relevant capabilities,” says a senior official from the Department of Science.

At the centre of this push is the ₹6,003 crore NQM, approved in 2023, which aims to build capabilities across quantum computing, communications, sensing, and advanced materials through 2031. The targets are ambitious: developing intermediate-scale quantum comput-